Analyst

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Cite this: *Analyst*, 2018, **143**, 4306

Investigation of the particle size-dependent near-infrared spectral features of binary mixture samples in conjunction with Monte Carlo simulation and the influence of particle size on the accuracy of quantitative analysis†

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Near-infrared (NIR) spectral features of a target component can be influenced by its particle size as well as the particle sizes of the surrounding components in multi-component samples; therefore, understanding the essence of particle size-induced spectral variation is fundamental for robust quantitative analysis. For systematic investigation, two different types of binary-component samples were prepared. First, in ambroxol/lactose powders, only the particle size of lactose (surrounding filler) was varied, while the particle size of ambroxol (a target component) remained unchanged. Second, in lactose/PE pellets, the particle size of lactose (a target component) was only varied. When measuring ambroxol/lactose powders, the absorbances of both ambroxol and lactose peaks were elevated as the particle size of lactose increased. The larger lactose particle made the photon propagation broader in the sample, so a greater amount of ambroxol was able to interact with NIR photons; this led to an increase in absorbance. In the case of lactose (10 wt%)/PE pellets, the absorbance of lactose oppositely decreased as the lactose particle size increased. Since PE was the dominant component (85–95%) and had a small particle size (7 μm), the depth of photon propagation was substantially shallow. In this situation, the larger lactose particles would sometimes not be located within the small sampling volume and/or the photons would only partially interact with the particles; this led to a decrease in the absorbance. For both binary samples, the accuracies for the determination of target concentrations deteriorated as the particle size increased. This was due to the degradation of the measurement reproducibility caused by the increased uncertainty in the photon propagation. Monte Carlo simulation was effective for probing the photon propagation inside samples and supportive to explain the experimental observations.

Received 21st April 2018,
Accepted 23rd July 2018

DOI: 10.1039/c8an00736e

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Introduction

Diffuse reflectance near-infrared (NIR) spectroscopy has been extensively used to analyze a wide range of solid samples, such as pharmaceutical tablets,[1–3] agricultural products,[4,5] and polymers.[6] Its ability to quickly measure samples in a non-destructive manner makes it useful for a wide range of real-field applications. Meanwhile, since NIR peaks originate from combination and overtone vibrations, they can easily overlap with each other, rendering the overall spectral features substantially less distinct. Therefore, multivariate analysis based on chemometrics[7] is typically adopted to extract relevant information from complex and overlapped NIR spectra. Under this situation, any unwanted spectral features irrelevant to the original identity of a sample, such as those caused by variations of particle size and instrumental drift, can adversely impact the accuracy of multivariate analysis. For this reason, these accuracy-degrading features should be minimized or eliminated.

Particle size is one major factor that influences the final NIR spectral features of solid samples; therefore, several previous studies have already reported the particle size-dependent spectral variations in the measurements of various samples, as summarized in Table 1.[8–13,15] These studies mostly reported that the absorbance of sample peaks decreased as the particle size became smaller. When a particle is small, the number of particles packed in a given volume is relatively large. Under this circumstance, NIR photons cross many particle-to-air interfaces during their propagation, and reflections occur at

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†Electronic supplementary information (ESI) available. See DOI: 10.1039/c8an00736e

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Table 1 List of previous studies that examined the NIR spectral features with various particle sizes and the observed trend of particle size-dependent absorbance change in each work

Sample	Type	Tested particle sizes (μm)	Trend[a]	Ref.
Lactose	Powder	74, 149, 210, 250	A	8
Wheat flour	Powder	150, 200, 250, 300	A	9
Sucrose	Powder	106–212, 250–310	A	10
Lactose	Powder	55, 105, 130, 180, 290	A	11
Phenacetin	Powder	41, 54, 76, 137, 254, 710	A	12
Microcrystalline cellulose	Powder	24, 45.8, 93.4, 261, 406	A	13
Lactose	Powder	44.7, 66.3, 98, 132, 168	A	13
Amoxicillin-trihydrate	Powder	Below 6.9, 11.2, 21.7	B	15

[a] A (particle size↑ → absorbance↑), B (particle size↑ → absorbance↓).

these interfaces. This causes the photons to diffuse over a relatively small space in a sample, resulting in a small sampling volume, thereby decreasing the absorbance. In a study of extracting scattering properties of turbid media with a single optical fiber, the higher absorbance was also observed with the longer photon pathlength, equivalent to the larger particle size.[14] In contrast, the opposite trend was found when the NIR spectra of amoxicillin-trihydrate powders with much smaller particles (below 21.7 μm) were examined.[15] It is necessary to note that all the investigations in the table were based on the measurements of single-component powders.

Many samples encountered in the actual field for NIR analysis are composed of multiple components, and the quantitative compositional analysis of these samples (*e.g.*, for the determination of active pharmaceutical ingredient (API) concentrations in pharmaceutical samples) has been extensively performed. When several components are mixed together in a sample, the propagation of photons becomes more complex, and the spectral features of a target component are influenced by its particle size as well as the particle sizes of surrounding components. This is an important issue that must be addressed for robust quantitative analysis since the alteration of the NIR absorption magnitude of a target component by the particle sizes of surrounding components will directly influence the accuracy. However, the profound study on this issue has been rarely reported.

For systematic investigation, we prepared two different types of binary-component samples in this study: powders of ambroxol/lactose and pellets of lactose/polyethylene (PE). In the ambroxol/lactose powders, the particle size of lactose changed between 23.9, 80.7, 119.7, and 197.0 μm, while the particle size of ambroxol was unchanged (20.3 μm). The concentration of ambroxol was varied from 9.0 to 16.0 wt% at a given lactose particle size in order to evaluate the particle size-dependent accuracy in determination of ambroxol content. These samples were designed to examine the spectral features of the target component (ambroxol) as the particle size of the dominant surrounding filler (lactose) was varied. In contrast, in the lactose/PE pellets, the particle size of lactose (a target component) varied between 37.7, 87.1, 156.5, 260.5, and 382.0 μm, while the particle size of surrounding PE was constant (7.0 μm). Also, the concentration of lactose changed from 5.0 to 15.0 wt% at a given lactose particle size. The adoption of pellet was to investigate the particle size-dependent spectral variation in a highly-condensed sample such as a pharmaceutical tablet. Similarly, the dependency of accuracy on the lactose particle size was also evaluated.

Initially, any systematic particle size-dependent trends in the diffuse reflectance NIR spectra of both samples were searched. Then, the partial least squares (PLS)[16,17] was used to determine the concentration of each target component in both cases, and the accuracies were evaluated in relation to particle size. To obtain supportive information to explain the particle size-induced spectral variation, the NIR spectra of pure lactose powders and pellets composed of different particle sizes were also collected, and subsequent variations of lactose peaks in the spectra were characterized. In all cases, Monte Carlo simulations[18,19] were employed to tract the photon migration in the samples and to explain the experimentally observed variation in absorbance according to particle size as well as subsequent PLS accuracy.

Experimental

Sample preparation

Pure ambroxol, lactose, and PE powders were purchased from Sigma-Aldrich. Lactose powders with different particle sizes were prepared by sieving the samples with sieves of appropriate sizes. The average particle sizes and corresponding standard deviations of the prepared samples in this study are summarized in Table 2. The particle sizes were determined using a particle size analyzer (Malvern Instruments, Malvern, UK). The lactose pellets were made by placing 500 mg of each powder in a stainless-steel die (diameter: 13 mm) and pressing the sample at a constant pressure of 5 tons for two minutes. The lactose/PE pellets (total amount: 360 mg) were prepared using the same procedure. The thicknesses of the pellets were approximately 2.7 mm.

Acquisition of diffuse reflectance NIR spectra of samples

NIR spectra of the samples were collected in diffuse reflectance mode using an FT-NIR spectrometer (ABB Bomem, Quebec, Canada) equipped with a quartz-halogen light source and an InAs detector. The angles of illumination and detection were

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Table 2 Average particle sizes of the samples employed in this study and the corresponding standard deviations

Sample	Designation	Particle size (μm)
Pure lactose powder	PO-1	37.7 ± 26.1
	PO-2	87.1 ± 35.8
	PO-3	156.5 ± 87.0
	PO-4	260.5 ± 118.5
	PO-5	382.0 ± 122.0
Pure lactose pellet	PE-1	37.7 ± 26.1
	PE-2	87.1 ± 35.8
	PE-3	156.5 ± 87.0
	PE-4	260.5 ± 118.5
	PE-5	382.0 ± 122.0

		β-Lactose	Ambroxol
Lactose/ambroxol mixture powder	AL-1	23.9 ± 15.9	
	AL-2	80.7 ± 45.2	20.3 ± 15.2
	AL-3	119.7 ± 79.9	
	AL-4	197.0 ± 84.8	

		β-Lactose	Polyethylene
Lactose/PE pellet	LP-1	37.7 ± 26.1	
	LP-2	87.1 ± 35.8	
	LP-3	156.5 ± 87.0	7.0 ± 5.0
	LP-4	260.5 ± 118.5	
	LP-5	382.0 ± 122.0	

45° with respect to the sample surface. The powder samples were transferred in a glass vial (diameter: 22 mm), and the vial was continuously rotated during spectral acquisition. NIR spectra of the powder samples were recorded as log(1/*R*) with respect to polytetrafluoroethylene (PTFE) powder (particle size: 27.8 ± 21.6 μm) as a reference. Each pellet was placed on an aluminum disk with a hole (diameter: 10 mm), and NIR radiation was illuminated through the hole for diffuse reflectance measurements. A pellet made of only PTFE powder was used as a reference. The resolutions for all of the collected spectra were 8 cm^{-1}. All spectral pre-processing and multivariate methods, including baseline correction, multiplicative scatter correction (MSC), partial least squares (PLS), and principal component analysis (PCA), were performed using MATLAB Version 7.0 (Math Works Inc., MA, USA).

Monte Carlo simulation

Monte Carlo simulations were executed using fully licensed Monte Carlo Version 2016. The basic assumptions for the simulations were that individual photons independently propagated through a solid medium in a random walk-like fashion in three-dimensional space, and incident radiation had a uniform intensity over the illumination area. The detailed parameters used for simulation, such as refractive index, scattering coefficient (μ_s), and absorption coefficient (μ_a), are summarized in Table S1.† The optical coefficients of lactose powder and pellet were obtained by a linear fitting based on the previous data acquired by radially diffused reflectance method.[11] Also, the same parameters were used for ambroxol, since it had similar optical property with lactose. According to the assumption obtained by near-infrared diffuse reflection spectra of PE in the previous study,[20] PE was set as a total reflecting particle with unity reflection, thereby zero scattering coefficient for the simulation. The number of illuminating NIR photons was 100 000 and the photons were evenly distributed over the illumination area in the simulation. The diameters of the illumination area and detection window were set to 10 and 30 mm, respectively, which are close to those of actual experimental configuration. After illumination of photons, the reflected photons, equivalent to the non-absorbed photons by a sample were counted at the detection window. Absorbance was calculated by $(\#_{\text{incident photon}} - \#_{\text{reflected photon}})/\#_{\text{incident photon}}$, where # indicates the number.

Results and discussion

NIR spectral features of pure lactose powders and pellets

Before examining the NIR spectra of binary-component samples, understanding the particle size-dependent spectral variations that occur in a single-component system is helpful. Therefore, the NIR spectra of pure lactose powders and pellets composed of particles with five different sizes were collected, as shown in Fig. 1. The lactose peaks in the 7163–3850 cm^{-1} range are displayed in the figure. The most noticeable observation in the spectra of the powders (Fig. 1(a)) is that the baselines of the spectra move up vertically as the particle size of lactose increases. Then, the lactose peaks are clearly present on top of the corresponding underlying baselines. To observe the variation in absorbance, the baselines of spectra were vertically offset to zero at 5042 cm^{-1}, and the absorbances of the strongest lactose bands at 4748 cm^{-1} (black circles) were

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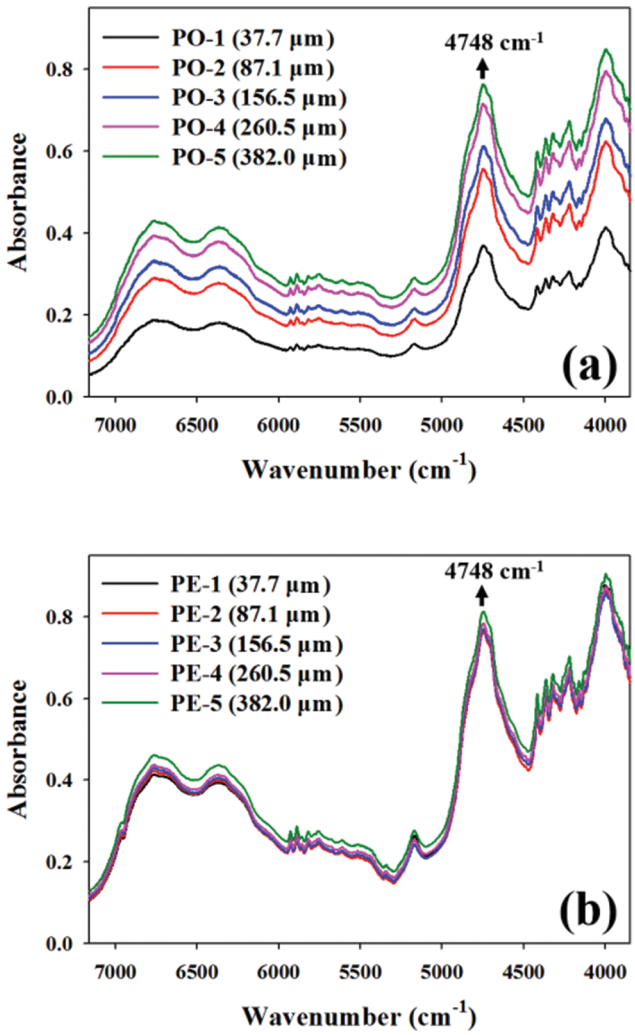

Fig. 1 NIR spectra of pure lactose powders (a) and pellets (b) composed of particles with five different sizes.

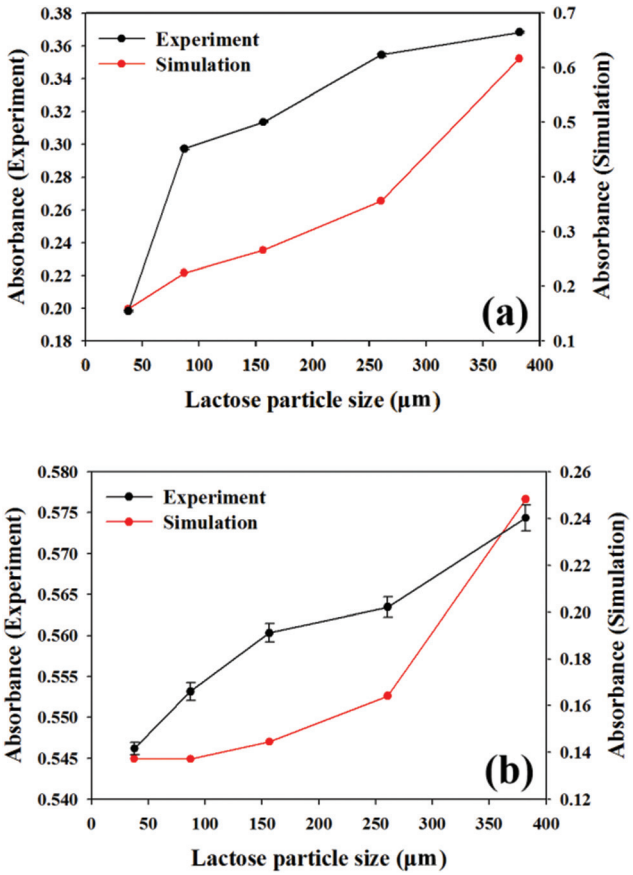

Fig. 2 Variation in absorbance of the 4748 cm^{-1} lactose band in the spectra of (a) lactose powders and (b) pellets. The black and red circles correspond to the experimental and simulation data, respectively.

plotted, as shown in Fig. 2(a). The error bar for each case corresponds to the standard deviation of six replicate measurements. For each measurement, the sample was repacked to ensure a random distribution of particles. The absorbance clearly elevates with increasing particle size. When a particle is large, it causes the photons to diffuse over a relatively large space in a sample, resulting in a greater sampling volume (as discussed in the Introduction). Therefore, this increase in absorbance was also observed in most of the previous publications (Table 1). Fig. 1(b) shows the NIR spectra of lactose pellets prepared using the lactose powders. Notably, the degree of vertical baseline variation among the spectra is substantially smaller compared with that in the spectra of lactose powders. The porosity in the pellet is much smaller, which means that the void volume is much less; therefore, the internal propagation of photons becomes more consistent, leading to smaller baseline variation. Fig. 2(b) shows the variation of absorbances at 4748 cm^{-1} after baseline correction (black circles). Again, the absorbance increases as the particle size becomes larger. The error bar in each case again corresponds to the standard deviation of six replicate measurements (three pellets were prepared for a given particle size, and both sides of each pellet were measured). To further investigate the effect of particle size on absorbance, Monte Carlo simulations were employed to track photon propagations in the lactose powders and pellets. Fig. 3 shows the distribution of reflected photons inside lactose powders consisting of five different particle sizes. The wavelength of 2106 nm corresponding to the 4748 cm^{-1} lactose peak was used for the simulation. The left and right plots in each case display the front and side views of the photon distribution, respectively. The dots indicate the locations where the photons are reflected. The angles of illumination and detection were 45° with respect to the sample surface, as previously stated. In the case of PO-1 (the smallest particle size), the photons are largely populated near the illumination area in the lateral direction, and the depth of distribution into the sample is shallow. Since the number of particle-to-air interfaces in a given volume is largest in this case, many photons are reflected in the region just below the packing surface as well as in the region near the illuminated area. As the particle size increases, the distribution of photons becomes greater in the lateral direction as well as deeper into

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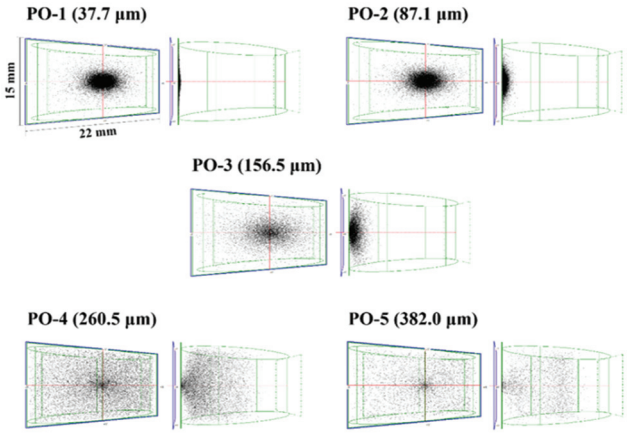

Fig. 3 Distribution of reflected photons inside lactose powders composed of particles with five different sizes. Left and right plots in each case display the front and side views of the photon distribution, respectively.

the sample. It should be noted that the photon distribution in PO-5 is substantially broader and deeper. The particle size of each sample varies in some extent as shown in Table 2 and the variation becomes greater when the particle size is larger such as PO-5 and PE-5; while, the nominal particle size was used for the simulation. So it is worthwhile to note that, the simulation would be relatively less representative of the sample when its particle size is larger due to the greater variation in the particle size.

Next, the number of photons arriving at the detecting window was counted, and the subsequent absorbance was calculated (red circles) relative to the total number of injected photons, as shown in Fig. 2(a). The trend of absorbance variation agrees with that observed in the experimental data. The elevated absorbance obtained with the larger particles is caused by the decreased number of photons reaching the detector (equivalent to attenuation of the intensity), from the reduced chances of photons to be detected due to their broader propagation inside the packed powders. The distribution of reflected photons inside lactose pellets composed of particles with five different sizes is shown in Fig. S1.† The red circle indicates the pellet (diameter: 13 mm). Again, the photon distribution becomes broader in the lateral direction as well as deeper as the particle size increases, which was also observed for the packed lactose samples. Meanwhile, the distribution of photons in the pellet is narrower since the interparticle distance is much shorter due to the compressed environment. Fig. 2(b) shows the simulation results for the lactose absorbance in the pellets (red circles). The increase in absorbance with increasing particle size was analogously predicted. In both cases (lactose powders and pellets), the simulation estimates the trends agreeing with the experimental observations. In contrast, Bittner *et al.* reported the opposite trend when measuring amoxicillin-trihydrate particles ranging in size from 6.9 to 21.7 μm; they observed a decrease in the absorption as the particle size increased.[15] However, these particles were much smaller than those employed in the studies listed in Table 1. To investigate this reversed trend, Monte Carlo simulations were again performed for lactose powders with smaller particles sizes (15, 26, 38, 50, and 75 μm). Fig. S2† shows the simulated absorbance values for these samples. For the three smallest particles sizes (*i.e.*, 15, 26, and 38 μm), the absorbance values were similar to each other. Then, the absorbance significantly increased when the particle size was larger than 50 μm. For deeper examination, the distributions of photons from the surface to inside with a 0.5 mm-thick interval were investigated as shown in Fig. S3.† The leftmost layer corresponds to the top of a sample. The color in each pixel indicates the number of photons at a given location. When the particle sizes are 15, 26, and 38 μm, the photons are localized at the first layer, and their lateral distributions are analogous. However, when the particle size is 50 μm, the photons start to penetrate the second layer, and the photon penetration becomes dramatically deeper in the sample composed of 75 μm particles. It seems that the diffusion of photons is more confined to the lateral direction when the particle size is smaller than 38 μm. Since the tested sample was lactose, these simulation results might not fully explain the reported opposite trend; however, it was clear that the behaviour of internal photon propagation was distinctly different when the particle size of a sample was relatively small (below 38 μm). This is worth investigating further in the future.

NIR spectral features of binary mixtures

In the case of a mixture, the NIR spectral features of a target component can be influenced by its particle size as well as by the particle sizes of surrounding components. For this evaluation, powder mixtures of ambroxol and lactose were initially investigated. In these samples, the particle size of lactose (a surrounding filler) was changed from 23.9, 80.7, 119.7, to 197.0 μm (designated as AL-1, Al-2, AL-3, and AL-4, respectively), while the size of ambroxol (a target component) was unchanged (20.3 μm). Fig. 4(a) shows the NIR spectra of pure ambroxol (20.3 μm) and lactose (119.7 μm) powders in the 7190–3910 cm^{-1} range. The absorption peaks of each component are distinct, and the spectral features overlap with each other in the mixture. Fig. 4(b) shows the NIR spectra of ambroxol/lactose powders (ambroxol concentration: 12.5 wt%) composed of lactose with four different particle sizes. The bands in the 5300–4800 cm^{-1} range, where the most intense ambroxol band at 5015 cm^{-1} is located, are highlighted in the inset. The baselines of the spectra increase as the particle size of lactose increases; this was also observed in the measurement of the single-component powder because lactose was the dominant constituent (87.5 wt%) in these samples.

Although the particle size of ambroxol is unchanged, the absorbances of both bands increase as the particle size of lactose increases, as shown in Fig. 5(a). This shows that the variation in the particle size of the surrounding lactose matrix clearly influences the absorbance of ambroxol. The increased lactose particle size broadens the photon propagation inside the sample, so a larger amount of ambroxol is able to interact

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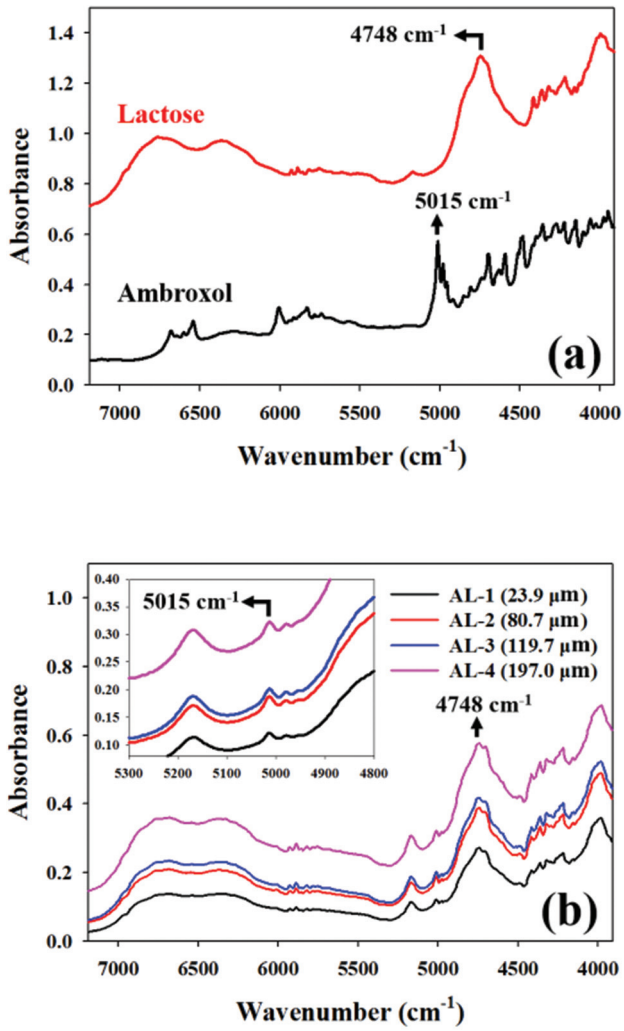

Fig. 4 NIR spectra (7190–3910 cm^{-1} range) of pure ambroxol and lactose powders (a), and ambroxol/lactose mixture powders (b) composed of four different particle sizes of lactose. The inset highlights the bands in the 5300–4800 cm^{-1} range.

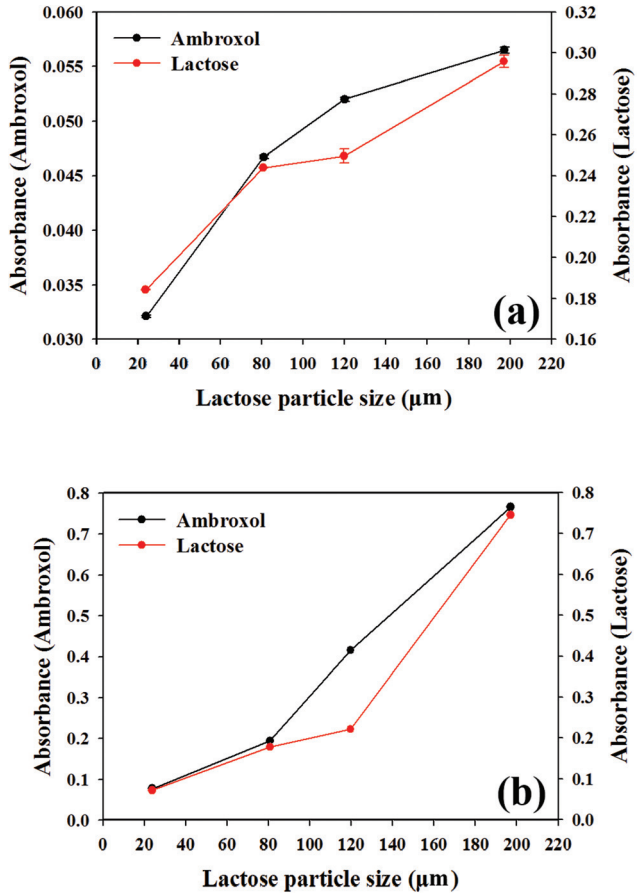

Fig. 5 Variations in the absorbance of the 5015 cm^{-1} ambroxol band as well as the 4748 cm^{-1} lactose band observed in the measurements of ambroxol/lactose powders composed of four different particle sizes of lactose (a) and the corresponding simulation data (b).

with the photons; this leads to an increase in the absorbance of ambroxol. The same reasoning can also be used to explain the increase in the absorbance of lactose. As also confirmed by the simulation shown in Fig. S4,† the distribution of ambroxol photons inside the ambroxol/lactose powders becomes broader when only the particle size of lactose increases. The distribution of lactose photons was also varied in the same fashion described above (relevant data not shown here), identical to the observation in Fig. 3. The simulation also predicted an increase in the absorbances of ambroxol and lactose as the size of the lactose particles became larger, as shown in Fig. 5(b).

Next, the accuracies for determination of ambroxol concentrations under the variation of particle sizes of surrounding lactose were assessed. At each lactose particle size, 15 samples with varying ambroxol concentrations (from 9.0 to 16.0 wt% at increments of 0.5 wt%) were prepared, and their diffuse reflectance NIR spectra were collected. Multiplicative scatter correction (MSC) was performed on the raw spectra, and the MSC-treated spectra were used for PLS to determine the ambroxol concentrations. Eleven randomly selected samples were used to build a PLS model, and the concentrations of the four remaining samples were predicted. This procedure was repeated 1365 times (corresponding to all possible combinations) to obtain the mean standard error of calibration (MSEC) and the mean standard error prediction (MSEP), as shown in Fig. 6(a). Two factors were used for PLS modeling in each case. In addition, the corresponding concentration correlation plots (actual *vs.* predicted concentrations) are shown in Fig. S5.† The black and red circles correspond to the calibration and validation data, respectively. As shown, the data become more scattered as the lactose particle size increases.

To determine the origin of the degraded accuracy with increasing lactose particle size, the reproducibility of the measurements in each case was analyzed. For this purpose, 12.5 wt% ambroxol samples with four different lactose particle sizes were selected, and six replicate measurements were performed for each sample. PCA was performed using all of the

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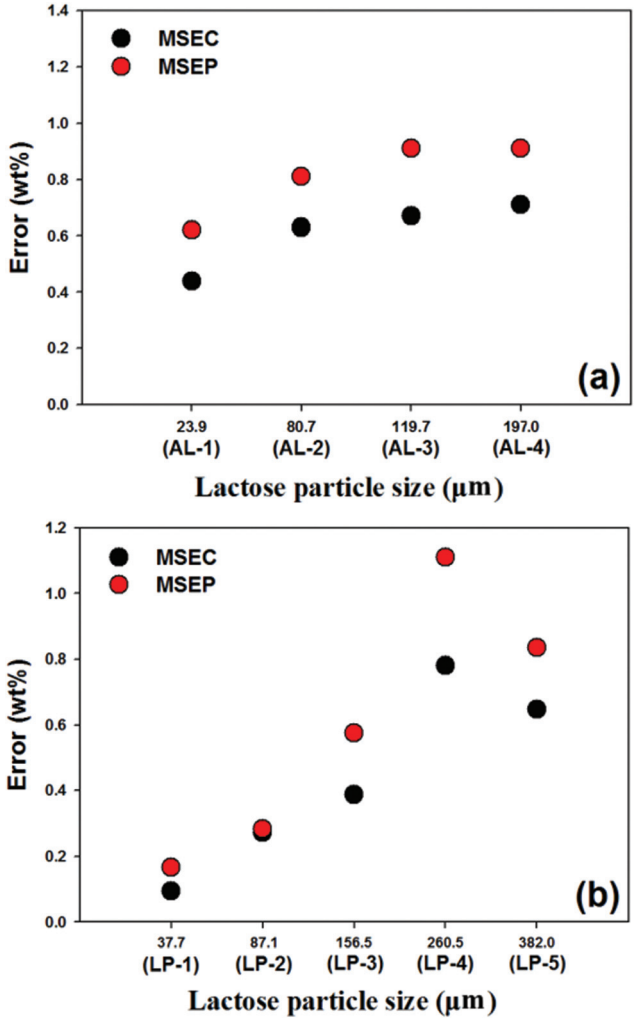

Fig. 6 MSEC and MSEP obtained in the determination of ambroxol concentration in ambroxol/lactose powders composed of lactose particles with four different sizes (a) and determination of lactose concentration in lactose/PE pellets composed of lactose particles with five different sizes (b).

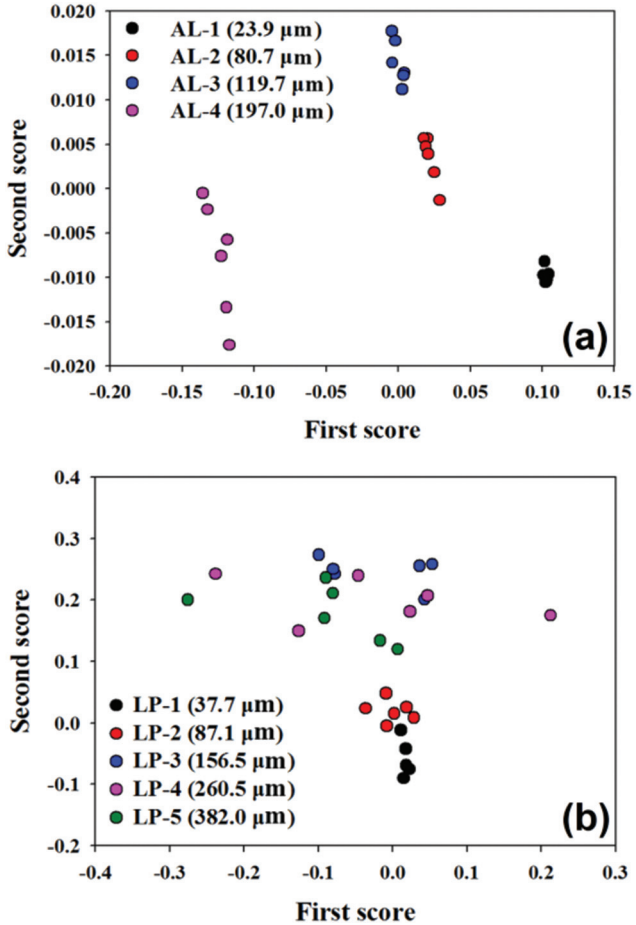

Fig. 7 PCA score scatter plots constructed using six replicate spectra of ambroxol/lactose powders (ambroxol concentration: 12.5 wt%) composed of lactose particles with four different sizes (a) and six replicate spectra collected from lactose/PE pellets (10.0 wt% lactose) composed of lactose particles with five different sizes (b).

MSC-treated spectra, and the score scatter plot constructed with the first and second scores was examined as shown in Fig. 7(a). The variation of scores became greater as lactose particle size increased, indicating the deterioration of reproducibility in spectral acquisition. Since larger particles cause photons to diffuse more broadly, the uncertainty in photon propagation is subsequently increased, leading to the degradation of reproducibility. Also, the simulation predicted that the number of photons counted at the detector became more variable (*i.e.*, a larger relative standard deviation (RSD)) as the lactose particle size increased when photons were separately injected into a sample 10 times (Fig. S6(a)†).

Next, pellets composed of lactose and PE powders were evaluated. The lactose/PE pellets were prepared by varying the particle sizes of lactose (a target component) between 37.7, 87.1, 156.5, 260.5, and 382.0 μm (designated as LP-1, LP-2, LP-3, LP-4, and LP-5, respectively) while maintaining the particle size of PE (a surrounding filler) at 7.0 μm. Fig. 8(a) shows the NIR spectra of pure lactose (382.0 μm) and PE (7.0 μm) pellets in the 7281–4235 cm^{-1} range. The absorption peaks of each component are clearly different due to their distinct molecular structures. Fig. 8(b) shows the NIR spectra of the lactose/PE pellets (lactose concentration: 10.0 wt%) where only the size of the lactose particle is varied. The 5030–4500 cm^{-1} range, which includes the lactose band at 4748 cm^{-1}, is magnified in the inset. The vertical baseline variation among the spectra is insignificant, since the lactose is a minor component, 10.0 wt%, in the pellet.

Fig. 9(a) shows the variations in absorbance of the 4748 cm^{-1} lactose band as well as of the 5665 cm^{-1} PE band. Unexpectedly, the absorbance of lactose decreased as the lactose particle size increased. Alternatively, the trend of absorbance variation for the PE band is unclear. It is worthwhile to note that the magnitude of absorbance variation of the PE band is very small: only 0.02 absorbance units. It is obvious that the absorption of PE can be influenced by the size of the lactose particles; however, the variation of the PE band is

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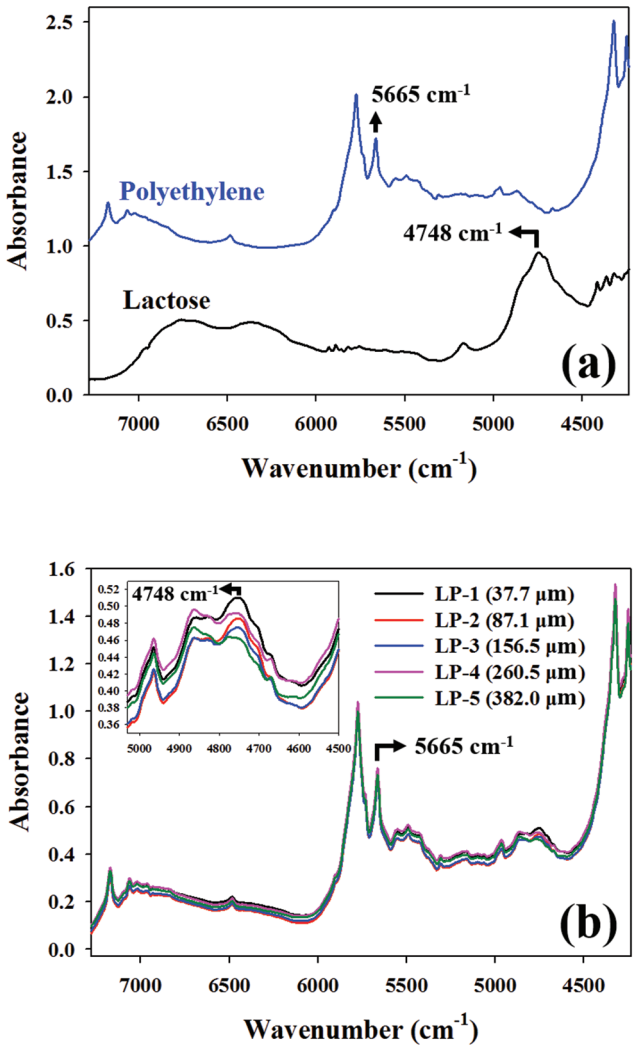

Fig. 8 NIR spectra (7281–4235 cm^{-1} range) of lactose and PE pellets (a), and lactose/PE pellets incorporating lactose particles with five different sizes (b). The inset highlights the peaks in the 5030–4500 cm^{-1} range.

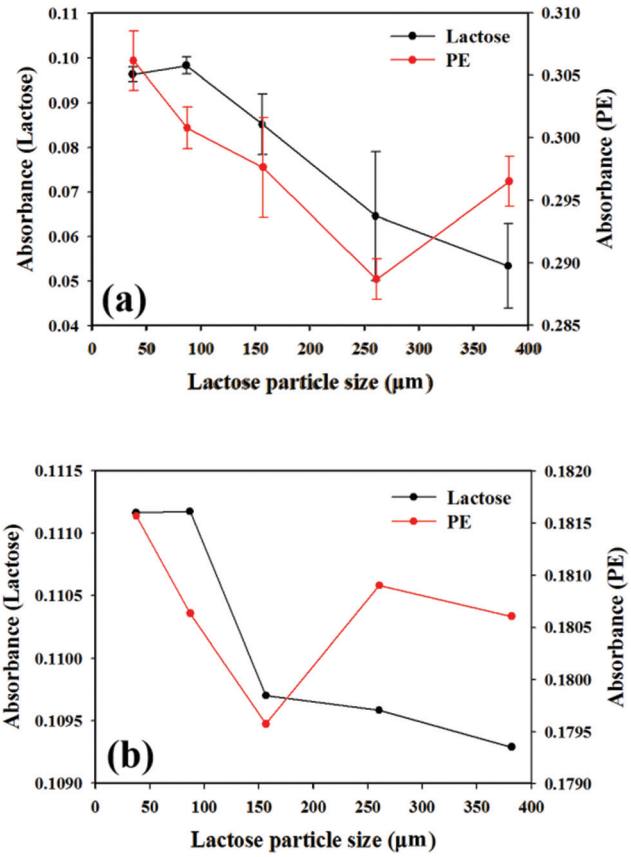

Fig. 9 Variations in the absorbance of the 4748 cm^{-1} lactose peak as well as in the 5665 cm^{-1} PE peak observed in the measurements of lactose/PE with five different particles sizes of lactose (a) and the corresponding simulation data (b).

minute, without any distinct trend because the lactose concentration is low. Meanwhile, the opposite trend observed in the absorbance variation of lactose bands still needs to be explained.

Fig. 10 shows the distributions of lactose photons inside the lactose/PE pellets. These photon distributions are similar for all five cases. Since PE is the dominant surrounding component (90 wt%) and has a constant particle size, the variation in lactose particle size did not significantly alter the overall photon distribution. In addition, the depths of photon distributions are substantially shallow (around 40 to 50 μm in depth based on the simulation) because the particle size of PE is very small (7.0 μm), and the PE particles are densely compressed. Meanwhile, the calculated number of lactose particles (10 wt%) in a volume of 1 mm^3 was 2189, 178, 31, 7, and 2 when its particle size was 37.7, 87.1, 156.5, 260.5, and 382.0 μm, respectively. The number of PE particles was 5 206 545. This calculation was based on the assumption that both lactose and PE particles are spherical. The number of lactose particles in the unit volume dramatically decreases as the particle size increases. In the 10.0 wt% lactose pellet composed of the largest particles, the number of lactose particles is expected to be only 228 (number of PE particles: 5.9×10^8); thus, the lactose particles are likely to be located far from each other. Under this situation, the lactose particles may not be present within the small sampling volume, and/or the photons may only partially interact with the particles. Therefore, the resulting absorbance could be smaller than those of samples with the same concentration but composed of smaller particles. The simulation also predicted a similar trend, as shown in Fig. 9(b).

The accuracies for the determination of lactose concentrations in the pellets were also evaluated. With a given lactose particle size, 11 samples with varying lactose concentrations (from 5.0 to 15.0 wt% at increments of 1.0 wt%) were prepared, and their diffuse reflectance NIR spectra were collected. Eight randomly selected samples were used to build a PLS model, and the three remaining samples were used for the validation. This procedure was repeated 165 times (all possible combinations for cross validation). Fig. 6(b) shows the resulting

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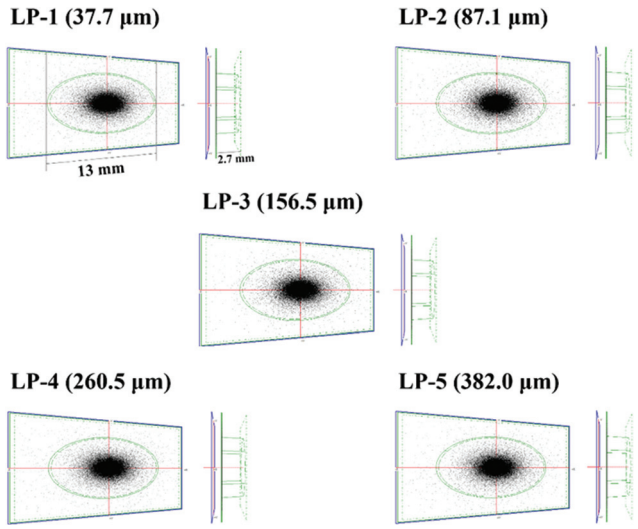

Fig. 10 Distribution of reflected lactose photons inside the lactose/PE pellets, in which the size of the lactose particle varies.

MSEC and MSEP for the determination of lactose concentrations in each case. The corresponding concentration correlation plots are also presented in Fig. S7.† Black and red circles correspond to the calibration and validation data, respectively.

Similarly, the accuracy becomes worse and the data points in the correlation plots are more scattered from the ideal line as lactose particle size increases. The score scatter plot obtained from the PCA of six replicate spectra of 10 wt% lactose pellets for each case (both sides of three separate pellets were measured) is shown in Fig. 7(b). Again, the scores became more scattered as lactose particle size increased, and the degraded reproducibility was mainly attributed to the increased error. The simulation also showed the same tendency of degrading reproducibility with increasing particle size (Fig. S6(b)†).

Conclusion

In the case of binary mixture samples, the NIR spectral features of a target component were influenced by its particle size as well as by the particle sizes of the surrounding components, and *vice versa*. The behaviour of particle size-dependent spectral variations can be influenced by many factors including the photon interaction (sampling) volume, particle sizes of individual components, and the particle concentrations. Monte Carlo simulations were particularly effective in probing the photon propagation inside the samples and supporting experimental observations. We are confident that this study is highly meaningful within the field of NIR spectroscopy because no previous profound or systematic investigations into the particle size-dependent NIR spectral variations for mixture samples have been reported. Although two different binary mixture samples of powders and pellets were examined, our findings can be referenced whenever robust NIR quantitative measurements of diverse mixture samples are necessary. A more widely acceptable strategy that maintains accurate quantitative analysis even under variation of particle sizes, either by effectively correcting particle size-induced spectral features or by using particle-size-adaptive experimental configurations for spectral acquisition, is currently in great demand. These issues are under investigation by our research group.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT & Future Planning (NRF-2018R1A2B2002662).

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